

LLD Case Study 9: Snake and Ladders Game

Target Patterns: Factory Pattern, Strategy Pattern, State Machine

Key Engineering Focus: Modular board layout, jump cell behaviors (Snakes vs Ladders), deterministic dice strategies, and multi-player game loops.

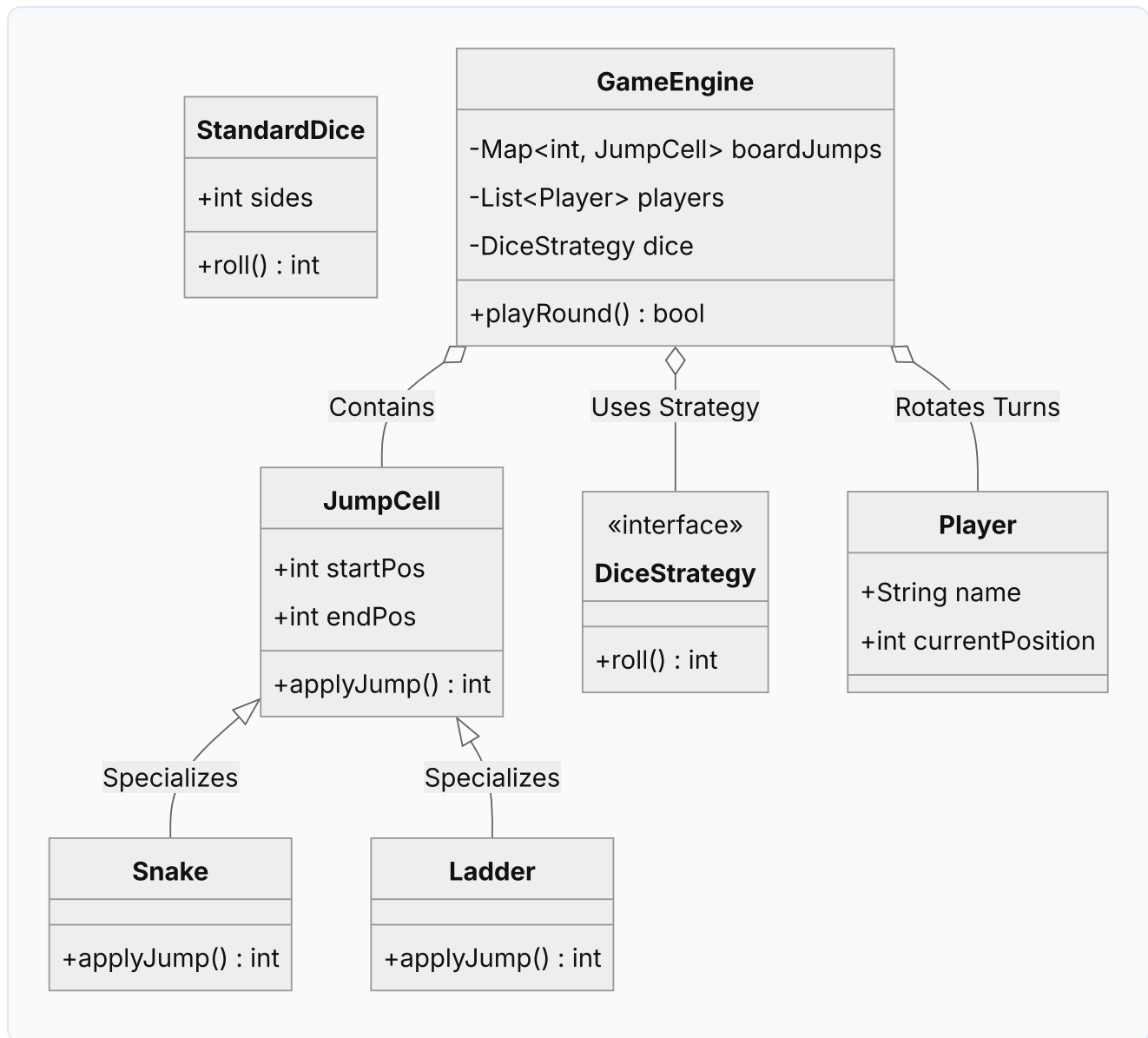
1. Problem Statement & Functional Requirements

Design a modular, extensible Snake and Ladders board game.

Requirements:

1. **Configurable Board:** Board with N cells (default 1 to 100).
 2. **Jump Entities:** Snakes (drops player back) and Ladders (propels player forward).
 3. **Dice Strategy (Strategy Pattern):** Pluggable dice mechanics (Single 6-sided dice, Biased/Rigged dice for testing, Dual dice).
 4. **Game State Engine (State Pattern):** Turns rotate in round-robin fashion until a player lands exactly on cell 100 to win.
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2. Architecture & UML Class Diagram



3. Production-Grade Python Implementation